Module 04 · Lecture 3

Introduction to HCI

Why HCI, how interfaces fail, and why we sketch

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Summary

- Why HCI?
- Introduction to HCI
- Getting started: how do interfaces fail?
- Sketching: attributes, sketch vs. prototype, the sketchbook

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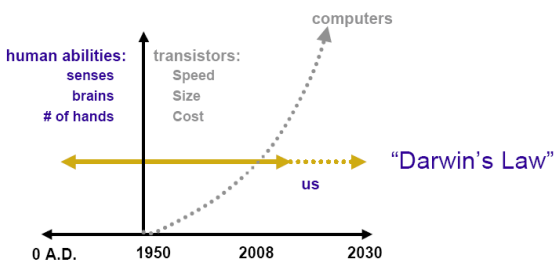
TOPIC 1

Why HCI?

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Machines improve exponentially. People don't.

Moore's Law:



Moore's law: speed, size and cost of computers improve exponentially.

"Darwin's law": human senses, brains and hands are the same as in year 0.

The gap between the two is where interaction design lives.

In 2026 add one more curve: interfaces generated by machines, faster than anyone can evaluate them.

Three more reasons, beyond the gap

The bottleneck moved

Computing is cheap and abundant. Human attention, time and understanding are not. The interface sets the ceiling of what all that computing is worth to a person.

Everybody is a user now

1965: a few thousand computers, with trained operators and manuals. 2026: over five billion people, no training, no manual. The design has to carry the knowledge.

Mistakes cost more

Screens now control cars, infusion pumps, power grids and dam gates. A misread number on a display is no longer a support call. It is an incident.

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Where the gap shows first: the data





Machines that feed other machines with data





Machines with new ways to interact

*Touch (2007), voice (2011 to 2014), headsets (2019, 2024), conversational and agentic interfaces (2022 to 2026) changed the **how**, not the **who**.*

Machines now produce most of the data and produce it for other machines.

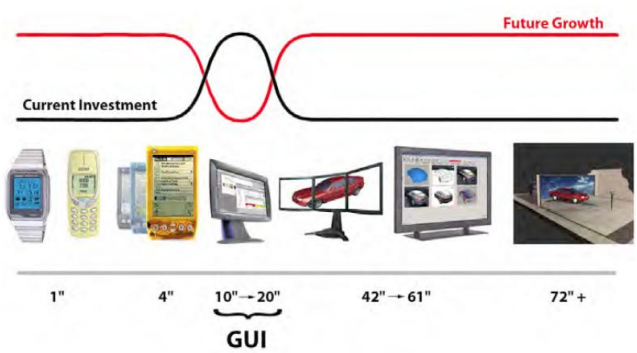
Each "V" of big data is, in fact, an interface question:

Volume	Millions of readings. Which few does the person need to see right now?
Velocity	New values every second. Redraw the screen, or only alert when something changes?
Variety	Numbers, maps, video, alarms. How do they share one screen without fighting each other?
Veracity	A sensor can fail. How does the display show how much to trust a value?
Variability	A normal day and a flood day. Should the same screen look the same in both?

Whatever the machines produce, a person still has to see it, understand it and act on it. That is the job.

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Investment in HCI: where the next growth comes from



Each curve is an S: a technology gives less and less for each extra euro, and the next curve starts with a new way of displaying and manipulating information.

The investment is already there: about half of the code and of the development time of a typical application goes into the user interface (Myers and Rosson, 1992).

Reading the curve

Current investment: the GUI. Since 1981: bigger screens, more pixels, flattening returns. A 72'' display does not double your productivity (but increases it).

Future growth: the next curve. Each new way to display and manipulate information opened a market that faster chips did not:

- **GUI and mouse (1984 to 1995):** computing reached every desk, not only the specialists.
- **Multi-touch (2007):** by 2011 smartphones outsold PCs.
- **Conversational interfaces (2022):** 100 million users in two months, no new hardware.

The next curve starts at the interface. That makes it an investment decision, not a finishing touch.

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TOPIC 2

Introduction to HCI

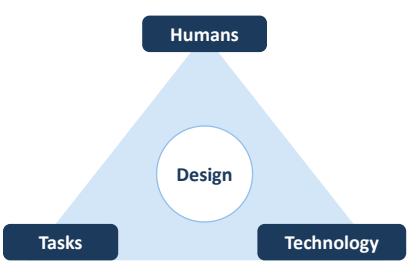
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What is user interface design?

Methods and theories for iteratively

- designing
- implementing
- evaluating

human interaction with a computer



One example you all know: the Multibanco (ATM)

Humans	Anyone from 15 to 85. In a hurry, on the street, in the sun or at night, with someone waiting behind them.
Tasks	Withdraw cash, pay a bill, check the balance. Rare, short, and a mistake costs real money.
Technology	A small screen, a keypad, a card slot, a cash slot. No mouse, no keyboard, no help desk.
Design	The choices that fit the three together: the words on the screen, the order of the steps, the beeps. Why does the card come back before the cash? Because people leave with what they came for, and forget the rest.
Iteratively	None of this was obvious when the first ATM opened in 1967. Each step was learned by watching people use it and fail.

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Some landmark HCI innovations

1963	Direct object manipulation (Sutherland, Sketchpad)
1965	The mouse (Engelbart)
1979	The spreadsheet (VisiCalc, Bricklin and Frankston)
1981	Desktop metaphor (Xerox Star)
1991	The Web and the browser
2007	iPhone: multi-touch as the main input
2010	iPad. Kinect: the body as controller
2011 to 2014	**Voice assistants (Siri, Alexa)**
2019, 2024	**VR and AR headsets (Quest, Vision Pro)**
2022	**Conversational interfaces with LLMs (ChatGPT)**
2025 to 2026	**Agentic interfaces: the system acts, the person supervises**

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HCI is interdisciplinary: who contributes?

The interface is where two families of disciplines meet. You come from the left column. Every design decision in the project borrows from the right one, whether you know it or not.

On the machine side

- Computer graphics
- Operating systems
- Programming languages
- Development environments
- Networking
- Software engineering

…and increasingly: industrial and product design, digital media processing, robotics

On the human side

- Art, graphic and tactile design: aesthetics (**layout, colour, icon selection, feel**…)
- Sociology and anthropology: group and **cultural behaviour**
- Psychology and kinesiology: **cognitive, perceptual and motor behaviour**
- **Human capabilities** to use and learn machines

Note: This course exists to cover the right column: the left one you already have covered.

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What makes it HCI?

What HCI is about	An example	…and how it fails
Joint performance of tasks by humans and machines	Sat-nav: the machine computes the route; the person drives and judges the road. Deciding who does what is the design.	The split fails when the machine keeps the easy part and leaves the person the hard one: an assistant steering for hours and expecting a takeover in two seconds. Watching a machine is what people do worst.
Communication: machine to human, and human to human through the machine	"Error 0x80070057" vs "The file name cannot contain /". Blue ticks: the machine tells someone that you read their message.	The machine speaks in codes and beeps. Blue ticks are a status, not a message, but they get read as one: "seen at 14:02, no reply". A technical state becomes a social statement, on by default.
Design methods: specifying, designing, implementing and evaluating interfaces	In this course: personas, sketches, a Figma prototype, cognitive walkthrough, heuristic evaluation, tests with real users.	The step that usually gets skipped is evaluation. "It works for me" is not a test: the developer is not the user.
Design trade-offs	Simple vs complete: a remote with 6 buttons or one with 60. Safe vs fast: "Are you sure?" before every delete.	Complete beats simple: the gym watch of Topic 3, eight functions on four buttons, and nobody finds the 'reset'. Safe beats fast: a warning before every delete, and everyone clicks OK without reading it.

Every row ends in a failure. That is Topic 3: how interfaces fail, and how to see it before your users do. (next slide)

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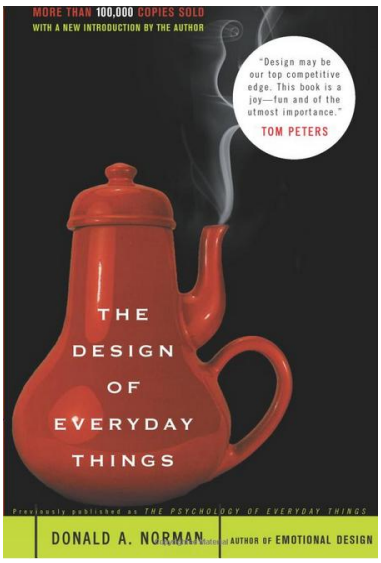
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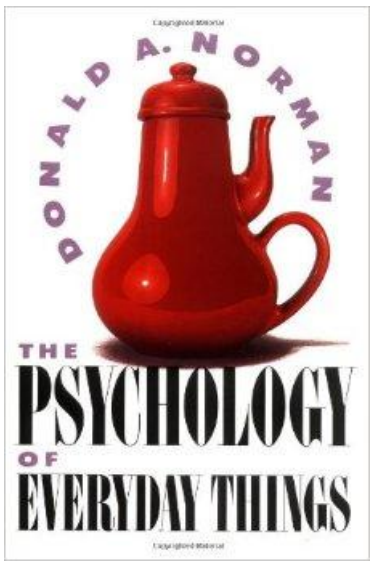
How do interfaces fail?

Getting started

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The Design of Everyday Things





Books by Don Norman

First published in 1988 as "The Psychology of Everyday Things", revised and expanded in 2013.

Obs: you may want to read this…

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The psychopathology of everyday things

Typical frustrations

- An engineer can't figure out how to heat two cups of milk with coffee in the company's microwave
- How many of you can program, or use all aspects of, your:
  - smart TV remote and streaming box
  - car's touchscreen
  - induction hob
  - washer and dryer
  - stereo system (home or car)
  - Excel worksheet
  - …



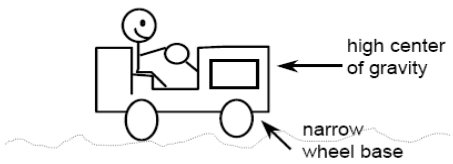
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Let's talk about early tractors

typical terrain:
un-surfaced
rough
hilly

original design:

early tractors



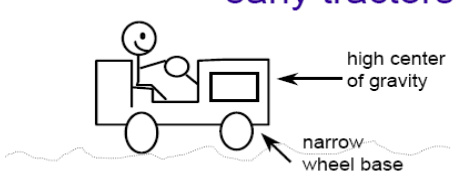
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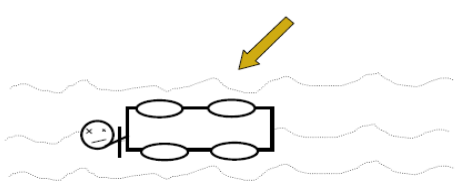
Let's talk about early tractors

typical terrain:
un-surfaced
rough
hilly

original design:

early tractors





used to be called "Driver's Error" …
but, accidents became infrequent when designs changed to low center of gravity & wider wheel bases

"Driver's error"?

Or a design that **ignored the terrain**, the centre of gravity and the wheelbase?
Lesson 1, in a tractor.

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The psychology of everyday things

Lesson 1: *the myth of human error*

Most failures of human-machine systems are due to:
- poor designs that don't recognise peoples' capabilities (+), and
- peoples' fallibilities (−)

This leads to apparent machine misuse and "human error"

Lesson 2

Good design accounts for human limitations.

Remember Moore's law vs. Darwin's law

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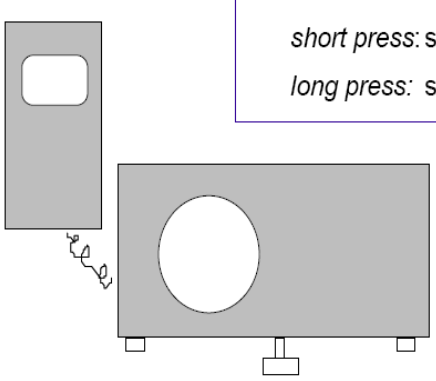
The remote control of the Leitz slide projector

how do you go forward/reverse?

instruction manual:

short press: slide change forward

long press: slide change backward



How do you go forward? And reverse?

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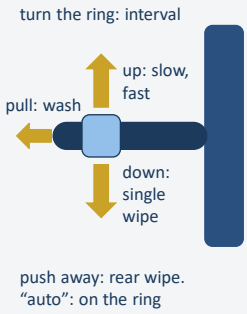
Hidden functions, then and now

The office phone, 1990s

- A number pad plus * and #. Dozens of functions hidden behind codes: *72 + number = forward all calls
- Can I remember the code?
- If I enter it, how do I know it caught?
- How do I remember that I left my calls forwarded?

The wiper stalk, 2026

- One lever, seven functions: single wipe, intermittent, slow, fast, wash, rear wipe, “auto”
- Can I remember which way is what? It changes between brands. In a Japanese car the stalks are swapped: you signal, and the wipers go
- Did I leave it on? “Auto” stays on, and the wipers start by themselves in the car wash



Ok, I will read the manual. Page 212, “Wiper and washer control”. Where is that manual anyway? Thirty years apart, the same failures: **functions you cannot see, codes you cannot remember, a state you cannot check**.

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Main concerns about HCI

Question for the class. From January 2026, Euro NCAP takes points from cars that put the indicators, the wipers, the horn, the hazard lights and the SOS call on a touchscreen. What went wrong there, in HCI terms?

Which information to present to the user?
In the car: speed and warnings, always. Tyre pressure, only when it matters.

Information

How to display or present that information?
A needle shows a change at a glance. A number has to be read. Red only for danger.

Display

How to manipulate and interact with the information?
A volume knob can be found by touch at 120 km/h. A menu on a touchscreen cannot.

Interaction

Check list: information, display, interaction. Use it for every panel in the coursework. Next slide: try it on a gym watch.

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The typical gym watch



Info: ?

Display: ?

Interaction: ?

…and only four buttons for interaction!

Functionalities

- Time: hours, mins, secs
- Date: year, month, day, weekday
- Light
- Chronograph: start, stop, reset
- Alarm: set time, on/off
- Time 1 and Time 2
- Adjust date (A/M/D)
- Adjust time (H:M)

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Psychology of everyday things

- Many so-called “human errors” and “machine misuse” are actually errors in design
- Designers help things work by providing a good conceptual model
- Designers decide on a set of users as the design audience (the personas)
- But **design is difficult**, for a variety of reasons that go beyond design!

- **You need to try and test several times!**





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“We want you to fail fast.”
What do you want with all this?
Why use bad drawings?
Why sketching?
Why not go for quality drawings?
Sketches make failing cheap!
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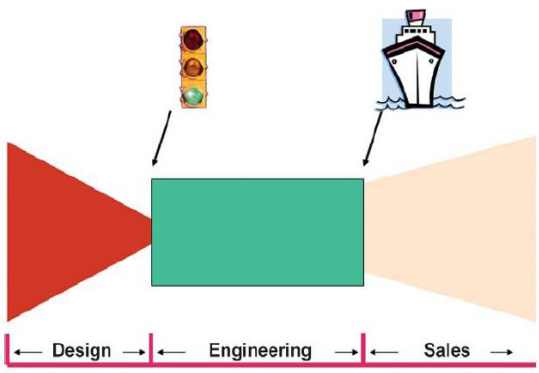
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Sketching
Attributes of a sketch, sketch vs. prototype, the sketchbook. Examples and motivation adapted from “Sketching User Experiences” (MK).
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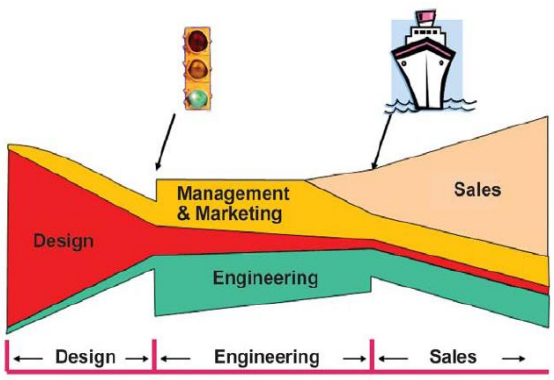
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The product view: integrating design

Many designs considered in parallel. Reduction filters and eliminates designs until convergence. One or more designs can be considered for the green light.

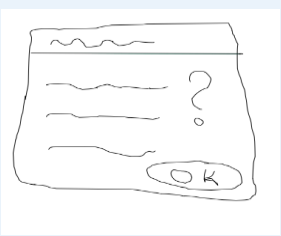


No barriers: actually, there should be an interplay between many different players (design, engineering, management and marketing, sales).



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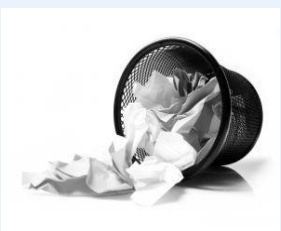
The attributes of sketches (1)



Quick
to make



Timely
provided when needed



Disposable
investment in the process and concept, not the execution. If you can't afford to throw it away, it's not a sketch. (Obs: in this course you shouldn't!)



Plentiful
they make sense in a collection or series of ideas: meaning and relevance in context

Form studies for a digital alarm clock: Baskinger, M. (2008) Pencils before Pixels. ACM Interactions, March+April, p. 32

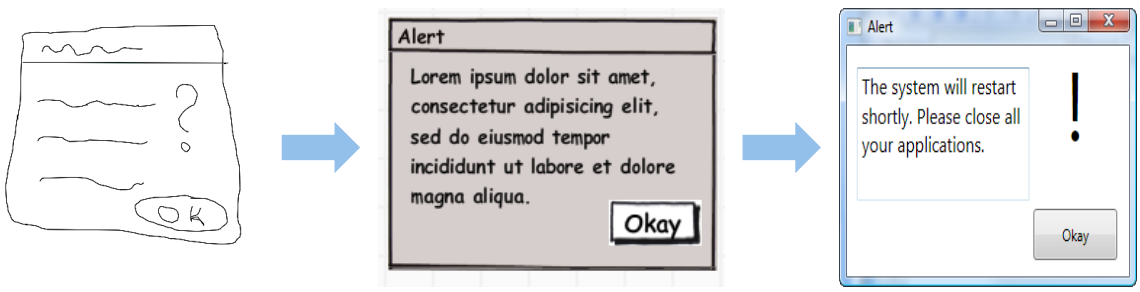
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The attributes of sketches (2)

Clear vocabulary
rendering and style indicate it's a sketch, not an implementation

Constrained resolution
no higher than required to capture its concept

Consistency with state
the refinement of the rendering matches the state of concept development



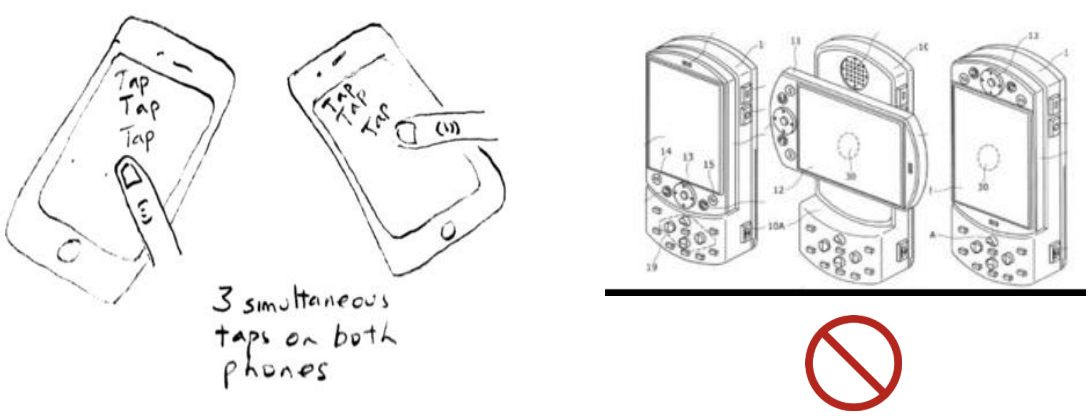
The same dialogue at three states of the concept: the rendering grows with the idea.

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Openness and freedom vs. tight and precise

Openness and freedom: incomplete, room to create

Tight and precise: complete, nothing left to do

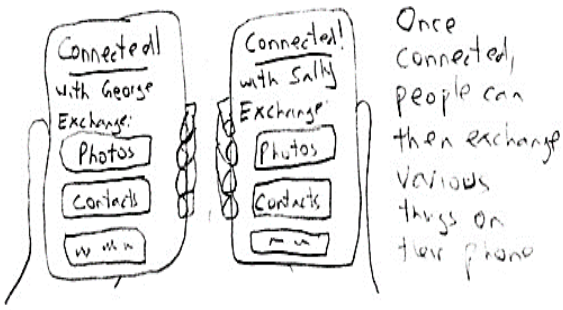


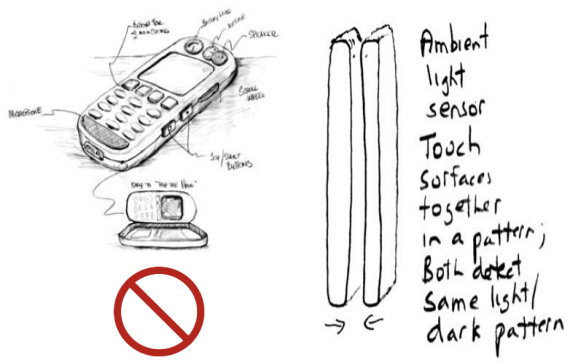
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Minimal detail, and refinement that matches the idea

Minimal detail: include only what is required to render the intended purpose or concept

Appropriate degree of refinement: make the sketch as refined as the idea. A solid idea, a more defined sketch. A generic idea, a less defined sketch.





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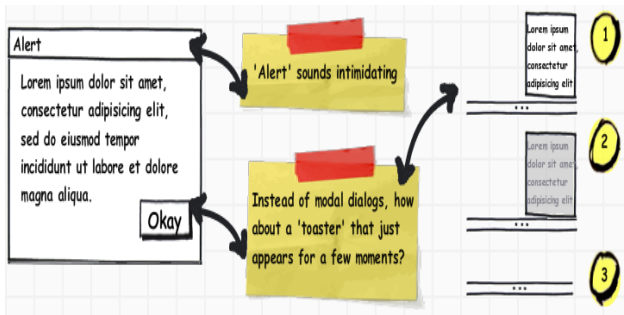
The attributes of sketches (3)

Suggest and explore rather than confirm

suggests and provokes what could be

A catalyst

evokes conversations and discussion



So far: quick, timely, disposable, plentiful, clear vocabulary, constrained resolution, consistency with state, suggest and explore, catalyst.

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A sketch is not a prototype

- **Difference:**
  - There is a contrast of purpose (always)
  - There is a contrast in form (usually, but not always)
- **But, it's a continuum**



sketch **prototype**

A sketch is for communication. A prototype is for verification.

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From sketches to prototypes

- **Sketches: early ideation in and at stages of design**
- **Prototypes: capturing and detailing the actual design**

Sketching is about UI design. Prototyping is about UI testing.

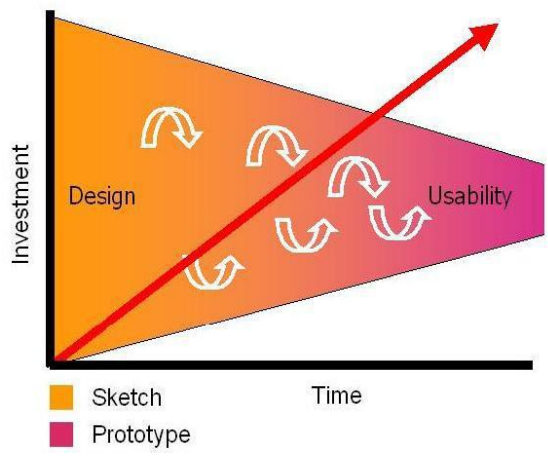


Image from Bill Buxton's book Sketching User Experiences (2007), Morgan Kaufmann

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From early design to late design

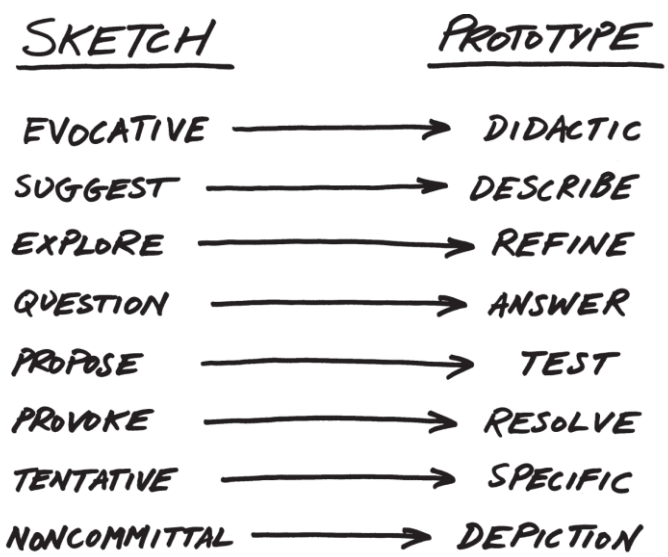
Early design

Brainstorm different ideas and representations	**Multitude of sketches**
Choose a representation. Rough out the interface style	**Sketch variations and details**
Task-centred walkthrough and redesign	**Sketches or low-fidelity prototypes**
Fine-tune the interface, screen design	**Low to medium-fidelity prototypes**
Heuristic evaluation and redesign. Usability testing and redesign	**High-fidelity prototypes**
Limited field testing. Alpha/Beta tests	**Working systems**

Late design

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Sketch properties vs. prototype properties



Sketch
evocative, suggest, explore, question, propose, provoke, tentative, noncommittal

Prototype
didactic, describe, refine, answer, test, resolve, specific, depiction

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A sketch vs. an AI-generated mockup

The sketch

- Quick, disposable, plentiful: ten in an hour, none precious
- Constrained resolution: it shows the idea, not the pixels
- Consistent with the state of the concept
- Suggests and provokes: people draw on top of it
- Made by you: you can explain every line

The generated mockup

- Looks finished: the wrong vocabulary for an open idea
- Resolution far above the state of the concept
- One polished answer instead of many different ones
- Closes the discussion: people react to the colours, not to the idea
- Not yours: the decisions inside it were made by nobody

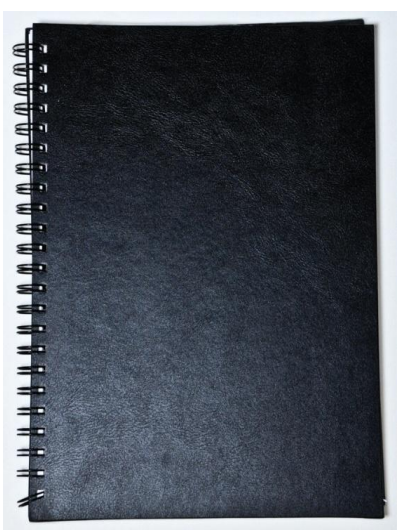
Course rule: AI-generated images are optional, and only to illustrate a concept the group has already chosen. Never as a source of ideas

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The sketchbook

Why a sketchbook?

- brainstorm many initial ideas, both good and bad
- explore and refine ideas, both in the large and in the small
- develop variations, alternatives, details
- reflect on changing thought processes over time
- communicate ideas to others by showing
- choose the ones worth developing
- record good ideas you see elsewhere
- clip inspiring images from sources like magazines
- shoot, print and collect inspiring photos



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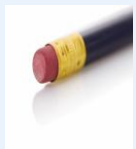
Sketchbook instruments



The pencil

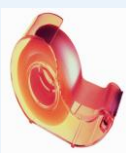
cheap, flexible, easy to carry (in the coil binding). Always carry one with you





The eraser and sharpener

handy, but optional





Tape, scissors and glue

for cutting and pasting in found objects. You don't have to carry them with you. Add your smartphone's camera!



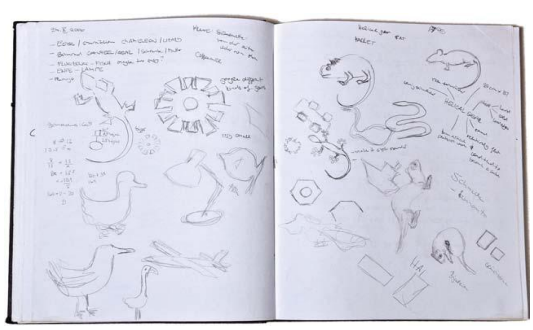
Other media

sure, as long as it doesn't get in the way

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Best practices

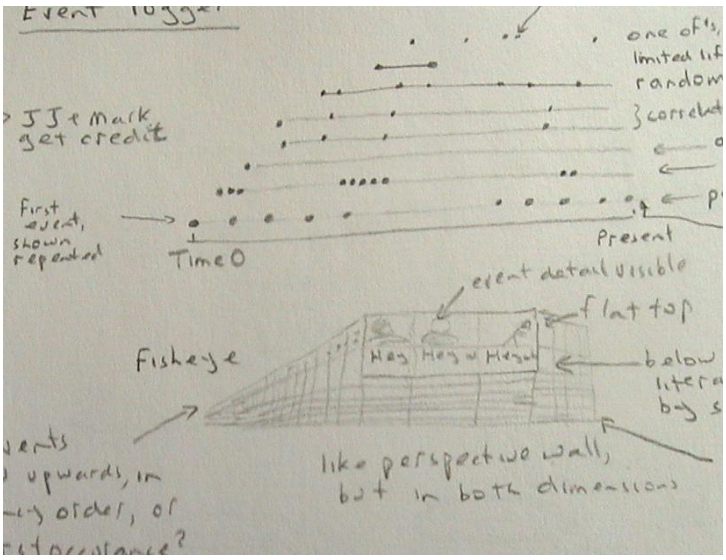
- **Carry and use your sketchbook regularly and frequently**
  - sketch anywhere, anytime. It only works if you carry it with you
- **Fill your pages**
  - a single well-composed idea, a series of related drawings, things you see that inspire ideas
- **Sketch alternatives (getting the right design)**
  - different aspects of the interface, different representations, different contexts of use…
- **Sketch details (getting the design right)**
  - detailed sketches that elaborate on interface nuances
- **Annotate drawings**
  - descriptions you cannot draw, textual addendums, sources of ideas, date…
- **Don't erase ideas. Don't use it for other things**
  - it's a record of all developing ideas, good and bad. For design only





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You don't need to be an artist!



An ugly, crude drawing can capture an idea as well as a beautiful one.

Portion of a page from Saul Greenberg's sketchbook

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Sketchbook examples: artists (not you!)



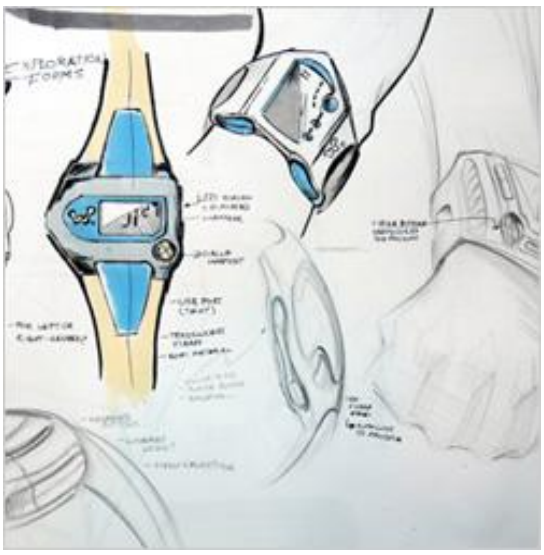


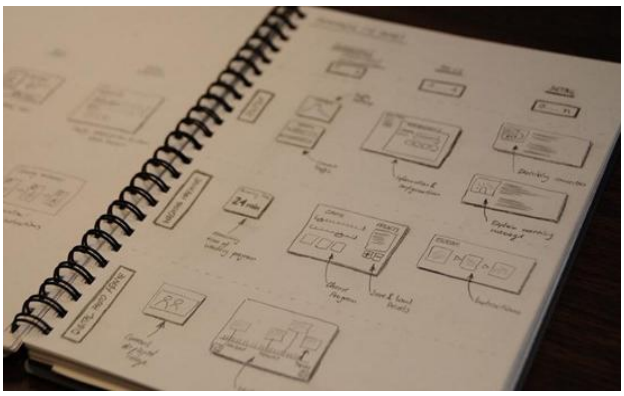
ideas, variations, studies

Sketch pages by Emily R. Feingold and from fmhs.cnyric.org (Studio Foundation)

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Sketchbook examples: ideas, variations, annotations

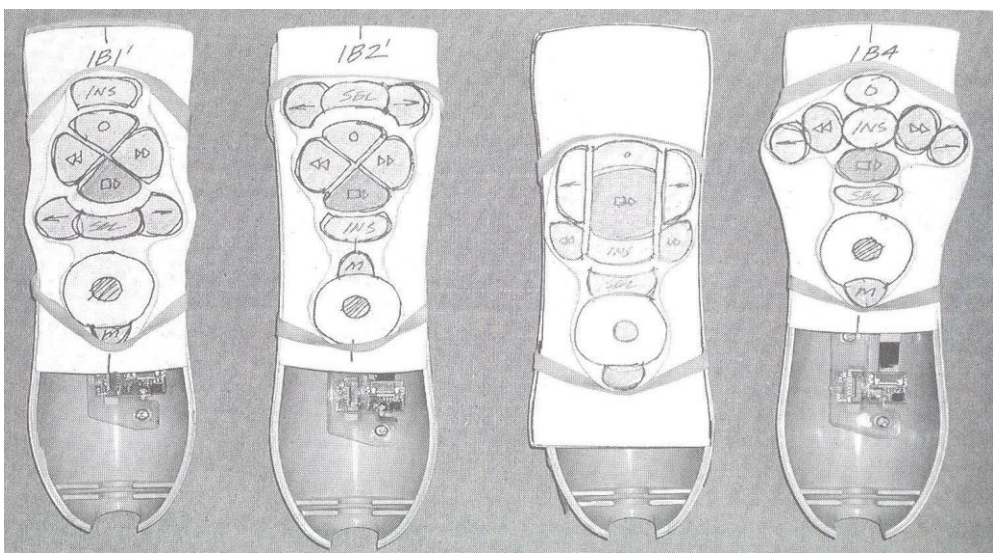




Left: form factors for a wearable computer watch (student industrial designer Samnang Eav). Right: four different versions of an idea (Nicolai Marquardt's sketchbook, with permission).

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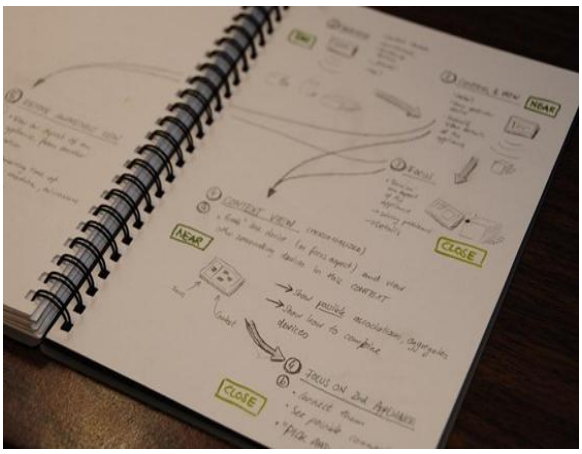
Sketch examples: design variations



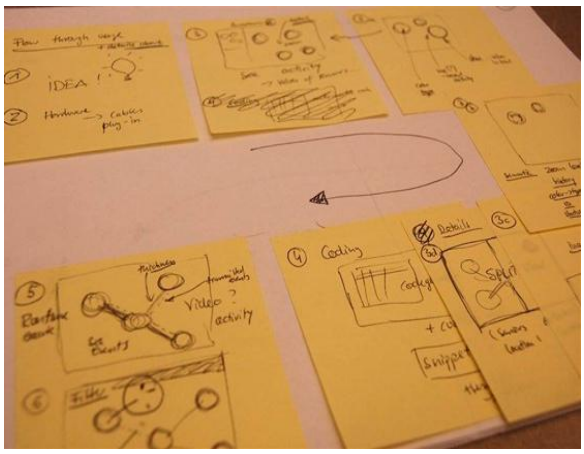
From Carolyn Snyder's book Paper Prototyping (2003), Morgan Kaufmann, p. 350

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Sketch examples: overviews, flow over time, storyboards



Overviews: arrows indicate relationships and flow

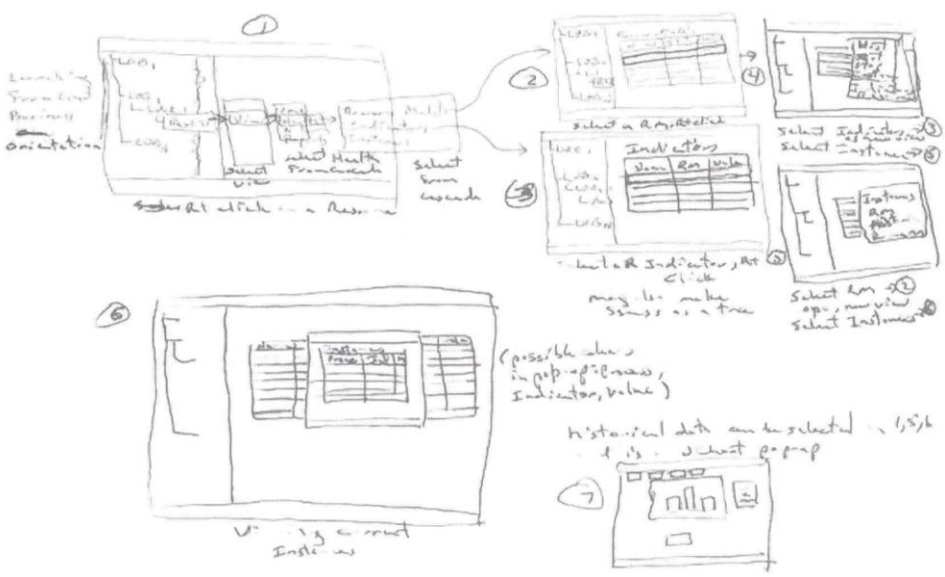


Storyboard: key scenes, like a comic book

Nicolai Marquardt's sketchbook, with permission

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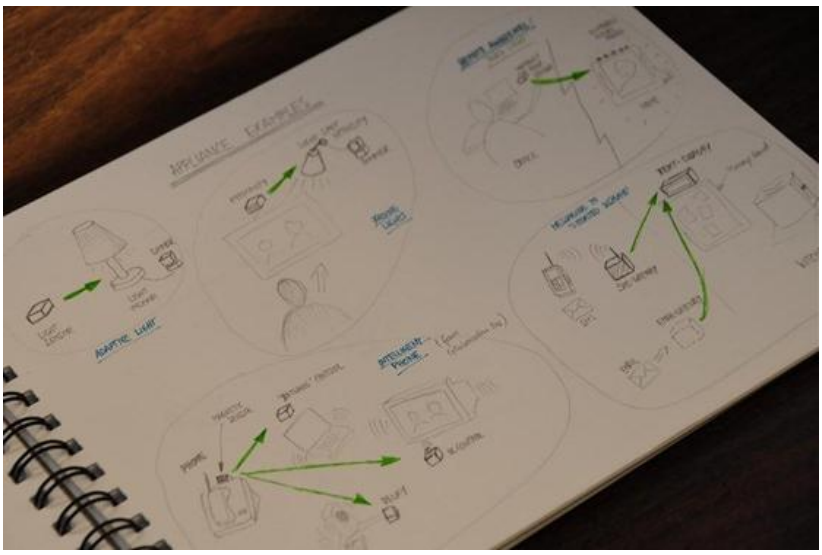
Sketch examples: storyboard transitions



From Carolyn Snyder's book Paper Prototyping (2003), Morgan Kaufmann, p. 11

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Sketchbook examples: scenarios



Scenarios

stories of envisaged uses: five envisaged scenarios of a technology in action

Nicolai Marquardt's sketchbook, with permission

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Wrap-up

Attributes of a sketch

quick, timely, disposable, plentiful, clear vocabulary, constrained resolution, consistent with the design state

A sketch is not a prototype

the difference is a contrast of purpose (always) and of form (mostly)

Sketch properties

evocative, suggest, explore, question, propose, provoke

Prototype properties

didactic, describe, refine, answer, test, resolve, specific

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